



Lessons from HART: Rail Wheel Problems and Solutions

Karl Kim, University of Hawaii

BOSTON



Overview

Background

Technical Challenges

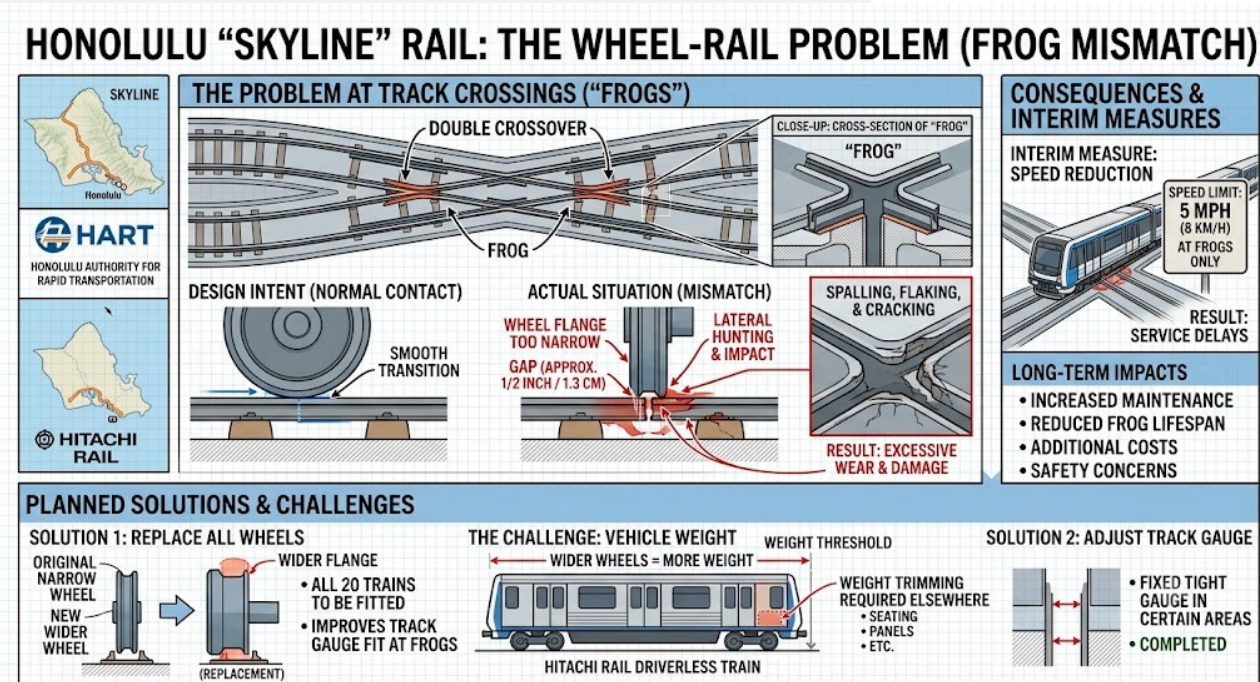
How did we get here?

Fixes and Failure(s)

WORST CASE SCENARIOS

Lessons and Takeaways

Future Opportunities





Consultant, Researcher, Professor

- Educated at MIT
- DTS/HART 1984
- OSP Study
- CCH Advisory Council
- Researcher
- Transportation Planning
- PSR-UTC
- FRA CRISI
- DHS FEMA NDPTC

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Supported by US DOT, FRA, CRISI

Railroad Industry Career Pathways

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Overview

- Address Workforce Needs
- Inventory, High School, Colleges, Professional Curricula
- Focus on Rail Operations, Maintenance, Management
- Safety, Engineering, Emergency Response
- Urban, Commuter, Regional, Short line, Class I

Programs and Approach

- Identify Job Categories, Skill Requirements
- NAICS and BLS data
- Hawaii's, California, and the rest of the nation
- Pathways for HS, Associate, Bachelor, Masters, PhDs
- Skills, Trades (Electrician, Welding, HVAC, Mechanics etc.)
- Disciplines: Engineering, Emergency management, Transportation Planning

Steps for Mapping Education-to-Career Pathways

Workforce Development Framework

Railroad Educational Requirements & Experience Levels

Education Requirements for Rail Careers (Total: 50 jobs)

Experience Levels in Rail Careers (Total: 50 jobs)

Hawaii Rail Industry Career Pathways Engagement Activities

Hitachi Rail Honolulu Maintenance Facility
Honolulu Rail Operations Control Center
Mechanics Externship, Leeward Community College

Faculty from Honolulu Community College tour railcar components and learn about advanced systems engineering in transit operations.
Highschool students and teachers observe real-time monitoring of rail operations, highlighting transportation technology and workforce needs.
Professor Bill Labby engages students and educators in hands-on mechanics activities, bridging classroom learning with rail and advanced manufacturing careers.

Moving Beyond Hawaii: Rail Industry Career Pathways

Job typing and Identifying job classes, Inventory and asses current relevant educational training programs, Identify Academic and Industry partners, Facilitate Engagement and review of data and information on job classification, skill requirements, and occupational needs. With students and industry partners, explore through field trips and site visits to rail industry facilities to experience first hand job and occupations. Working with universities and industry partners, develop curriculum to address skills and knowledge gaps of current and future rail industry employees. Identify training needs for skill trades people and professionals engaged in railway operations, maintenance, emergency response, security, communications, system monitoring, cyber, fare collection, capital program management, intergovernmental relations, and governance.



Editorial | Island Voices

Five fixes could help put Honolulu's rail back on track

By Karl Kim
Posted June 22, 2016
June 22, 2016

Karl Kim, Ph.D., is professor of urban and regional planning at the University of Hawaii, where he is conducting research on averting disasters and building resilient communities.

As a longtime observer and frequent commentator on Honolulu's rail project, I have come to expect setbacks and shakeups as the system creeps toward completion.

Yet the recent news of it stalling out at Middle Street suggests that we need change in leadership, ownership and commitment or surely the project is destined for failure.





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Aka Dr. Disaster!



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THE MAGAZINE OF HAWAIIAN AIRLINES

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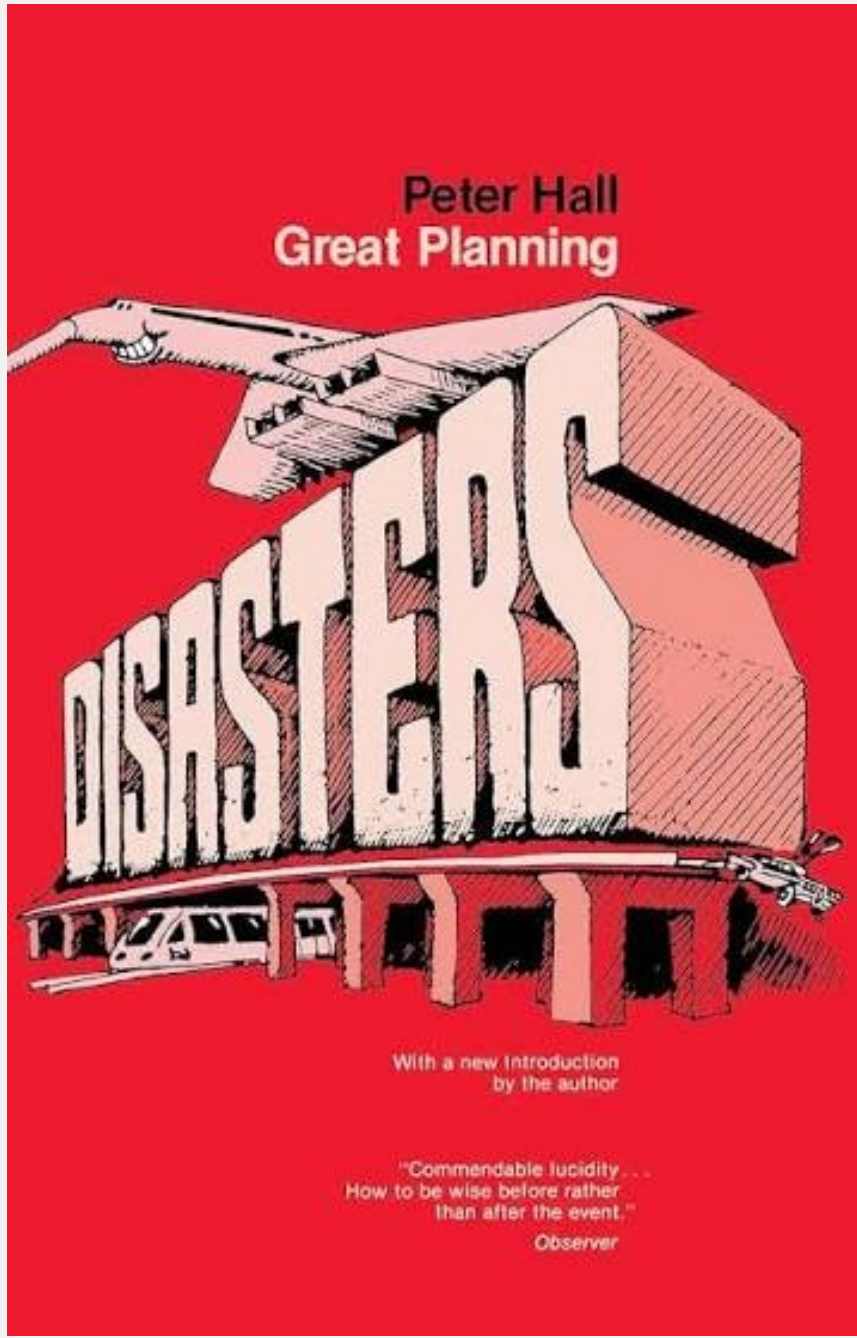


DEPARTMENT

Dr. Disaster

Hawai`i's own Karl Kim has the Rx for surviving earthquakes, eruptions, floods and fires

Story by Curt Saphurn



What went wrong?

Fixes

Short term

LONG RANGE

\$\$\$

What else?

What if?

Scenario Planning

B O S T O



NCHRP 20-44 (57) Implementing Scenario Planning

CALLING STATE DOT STAFF & LEADERSHIP
TELL US HOW YOU PLAN
FOR UNCERTAINTY



MAKING SCENARIO PLANNING WORK IN TRANSPORTATION

The **NCHRP 20-44(57): Implementing Scenario Planning Across Transportation Agencies** study is aiming to make scenario planning easier to apply, repeat, and integrate into everyday transportation work.

Be part of the conversation shaping how transportation agencies plan, prioritize, and act under uncertainty. Scan the QR code to **take the 5-minute survey or join a short follow-up conversation**. Prior experience with scenario planning is not required.



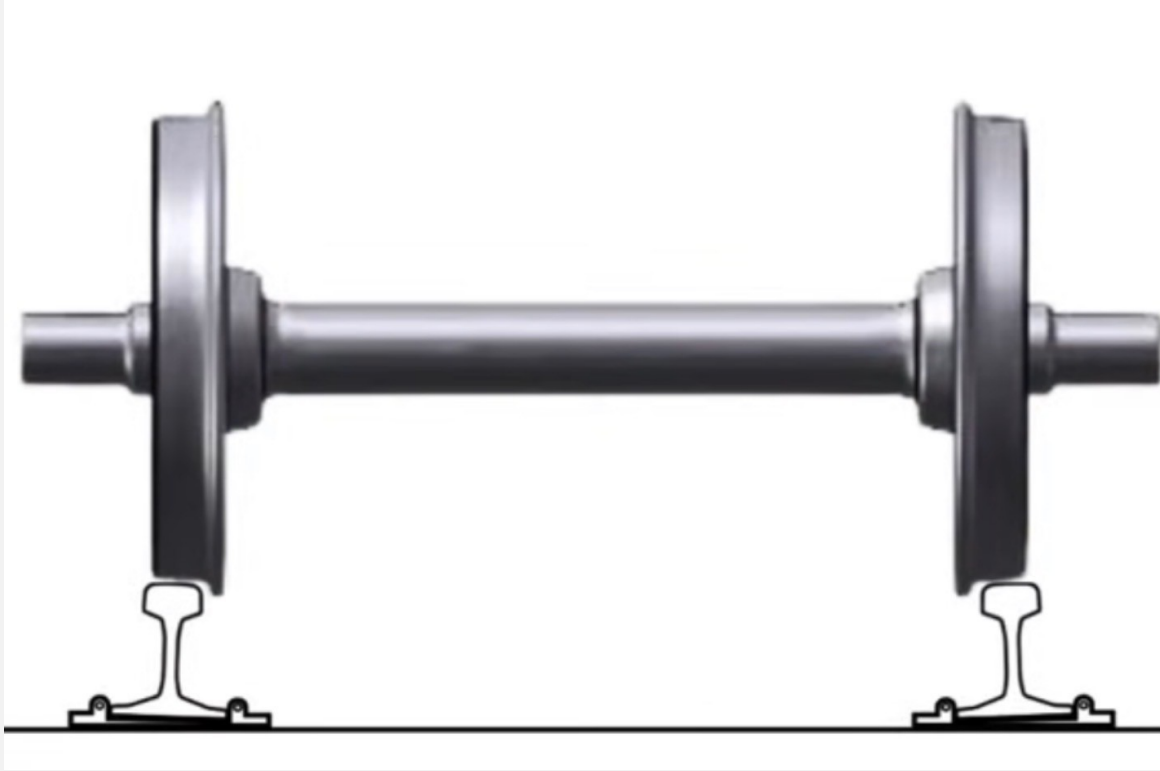
Dewberry®



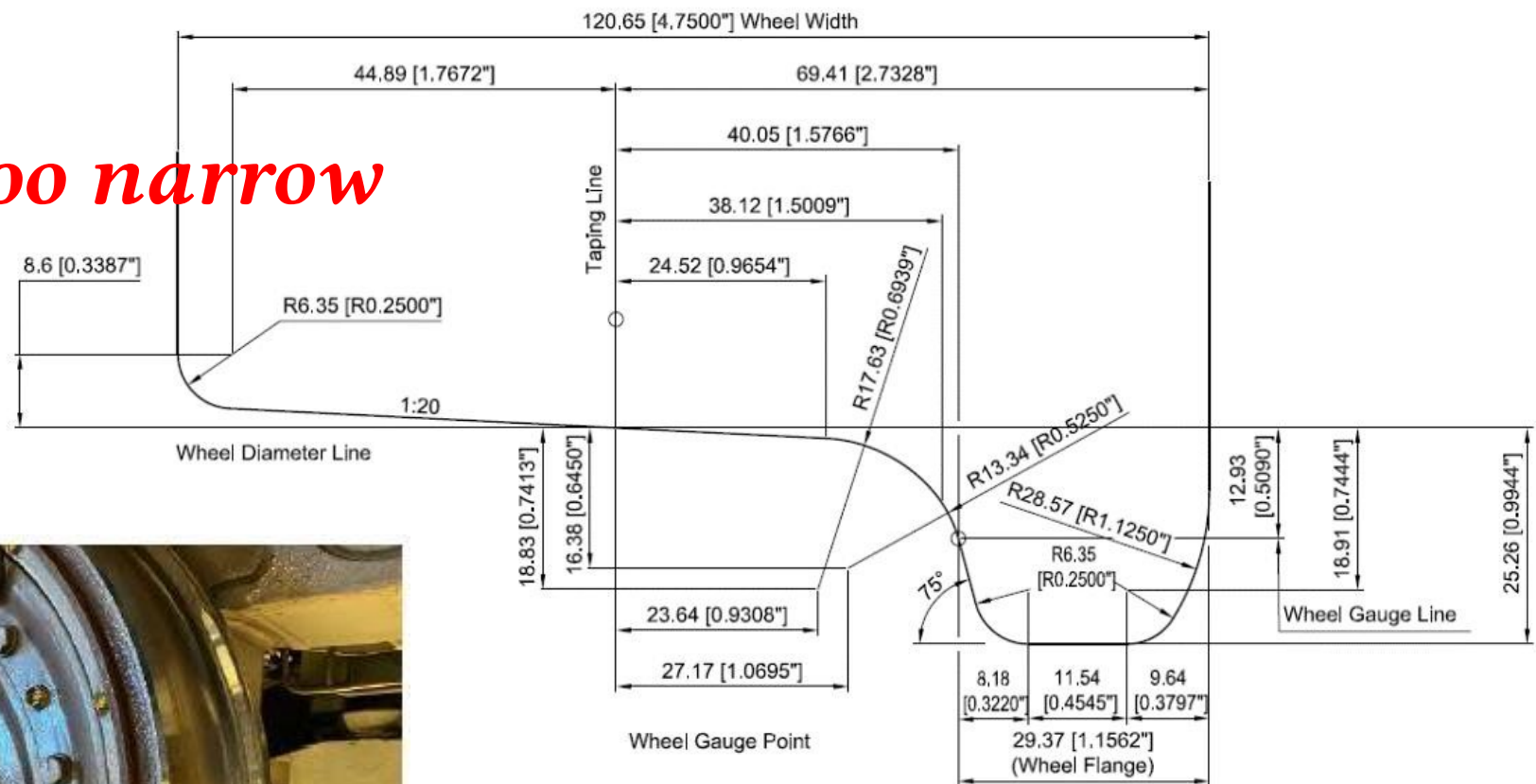
Elevated, driverless, automated, 19 stations, park n ride, TOD...



WWW (*what went wrong*)?



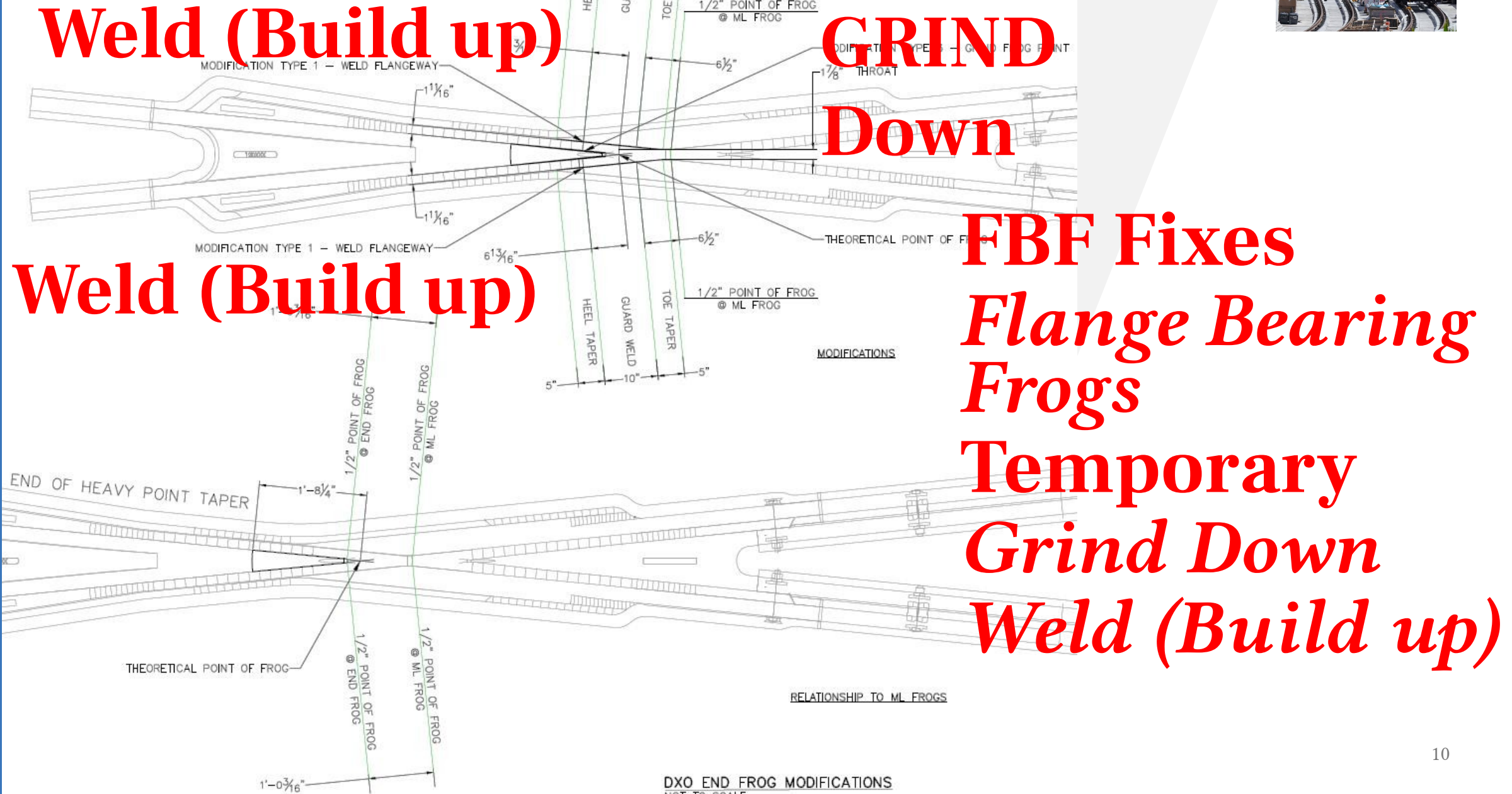
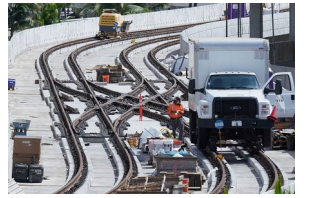
Wheel is too narrow



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THE MEASURES BETWEEN [...] ARE IN INCHES

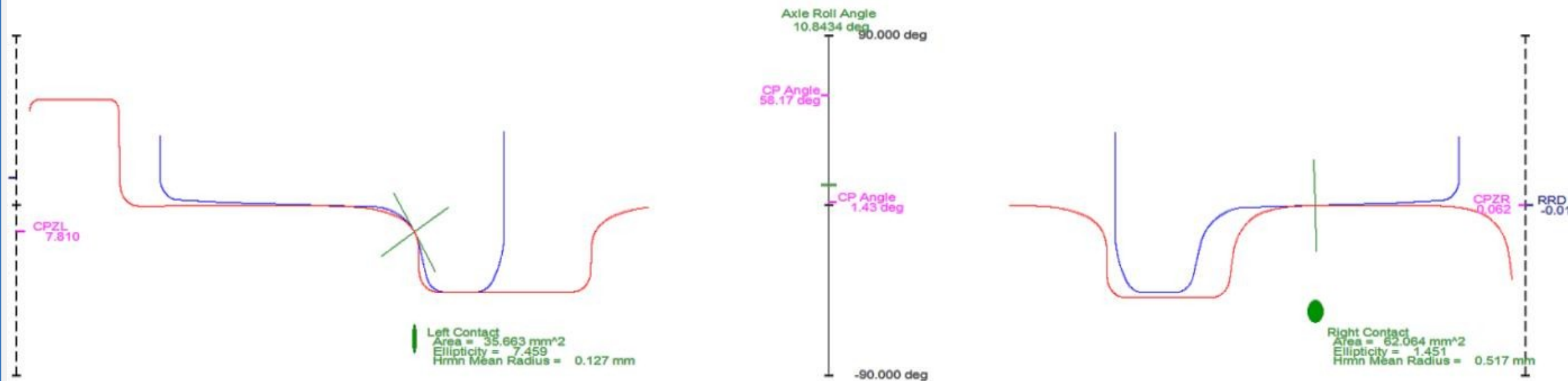
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		MASS TRANSIT Metro Honolulu Special wheel profile for running test	
		DYN-1873-D	00 1 / 1

ith logo





Longer term fix: Replace with Bigger Wheels

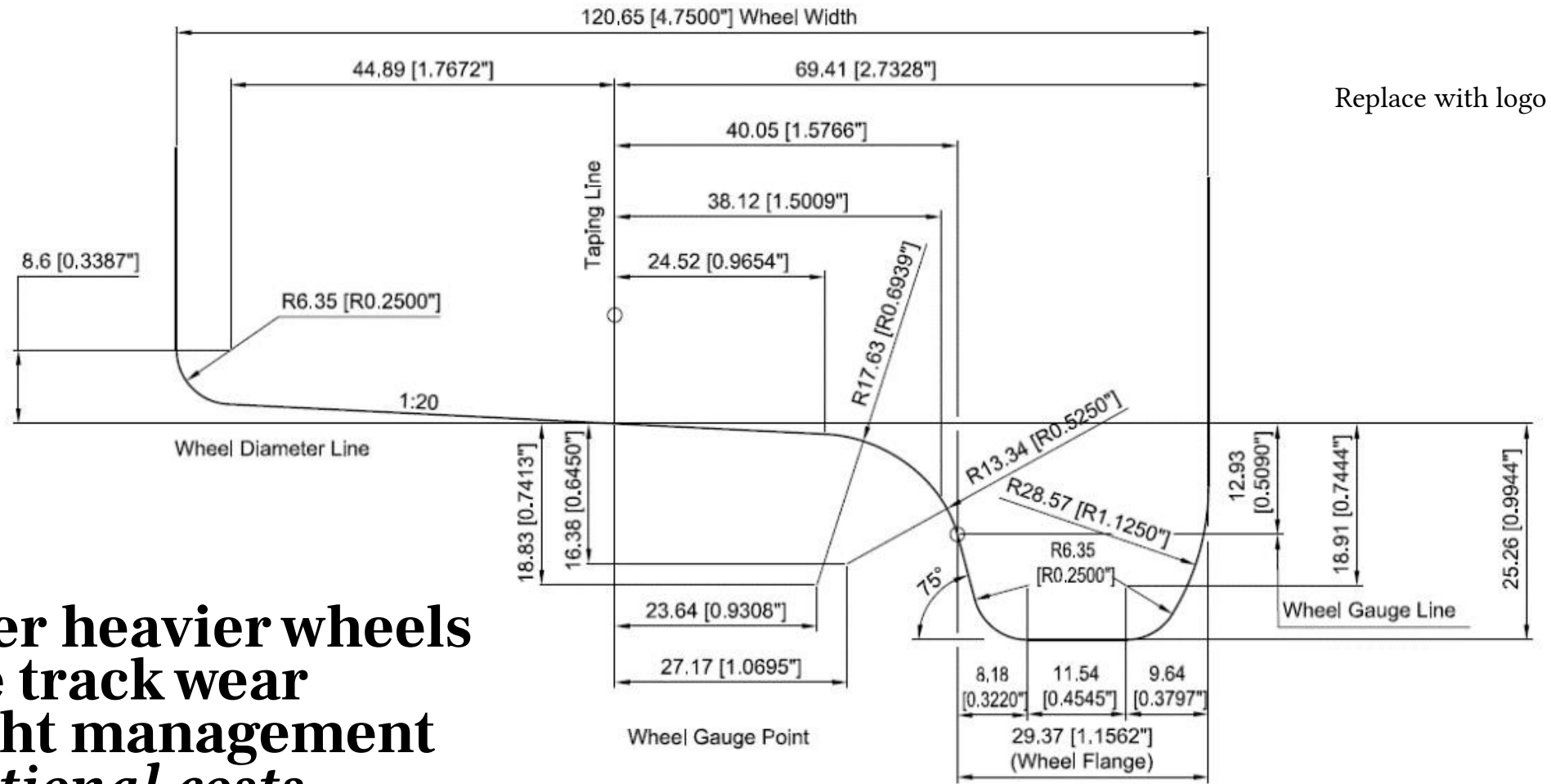


4.75" ---> 5.28" - 5.44"





Bigger heavier wheels
More track wear
Weight management
Additional costs
Operations
Speed
Monitoring
Risk management



Replace with logo

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THE MEASURES BETWEEN [...] ARE IN INCHES

Geometrical Tolerances ISO 2768 - mK ISO 13920 - BF			
Mass	kg	Department	Revision Description
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HITACHI Hitachi Rail SpA		DYN-1873-D	
		00	1 / 1



Initial risk assessment Modeling and Simulation Field Testing APTA AAR Guidance

FLANGE BEARING FROGS: TESTING, MODELING, AND MODIFICATION REVIEW

for Honolulu Authority for
Rapid Transportation/Lea+Elliott, Inc.

P-21-036

October 22, 2021

PROPRIETARY REPORT



NUCARS® Applications

- ◆ **Analyses of vehicle/track dynamic interaction**
 - Development of new vehicles and track components
 - Vehicle certification
 - Analysis of performance problems
- ◆ **Derailment analysis and investigation of derailment mechanisms**
- ◆ **Parametric studies**
- ◆ **Evaluation of specific vehicle and track design details and defects**
- ◆ **Evaluation of wheel/rail profiles**
 - Wear and RCF
 - Switches and crossings
 - Design optimization
- ◆ **Operational and environmental conditions**
 - Vehicle Ride Quality
 - Wheel-Rail Lubrication Studies
 - Dynamic Clearance Envelope Calculations
- ◆ **Tool for learning details of vehicle/track interaction**

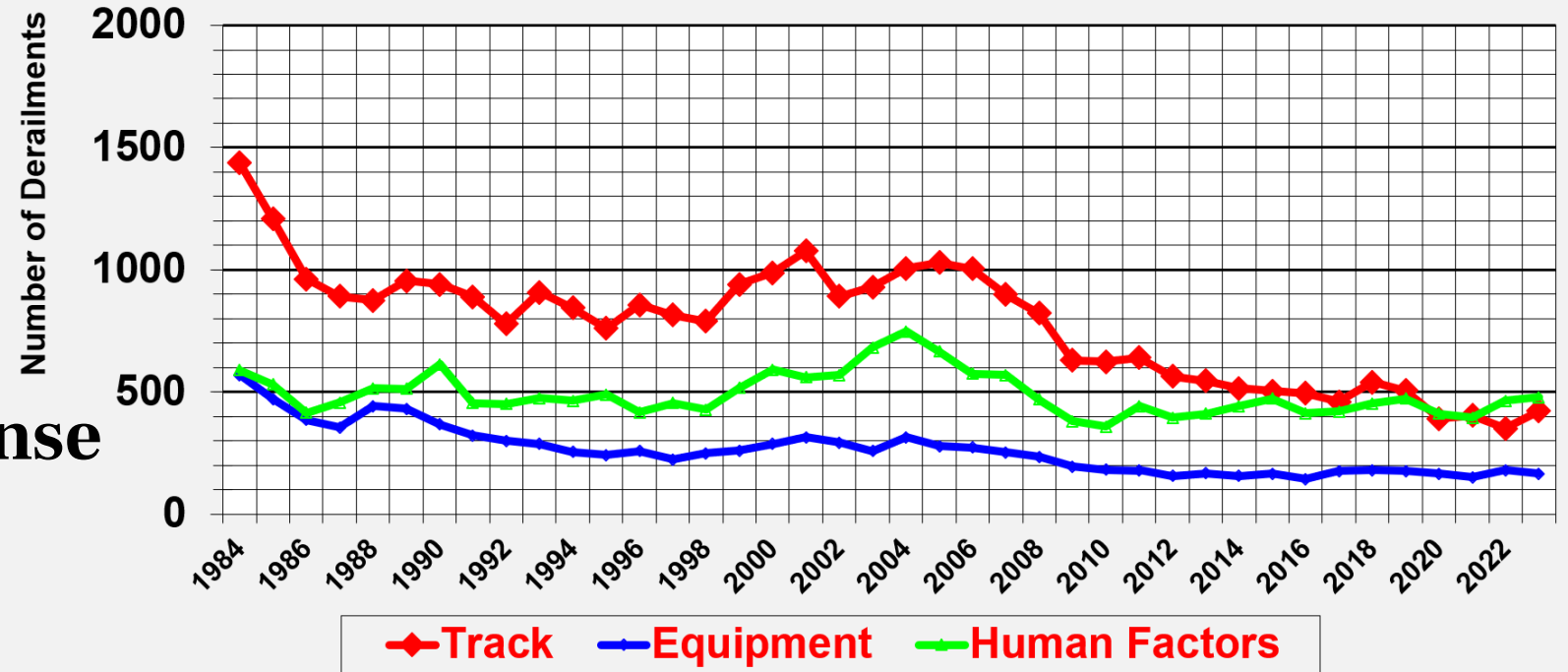
BOSTON

Mahalo to MxV!!!

Risk-based *Scenario Planning*

- **Derailment**
- Fully loaded train
- **Peak time**
- Congestion
- **Emergency Response**
- Detection/Alert
- *Unique challenges*

1984-2023 FRA Reportable Derailments by Cause



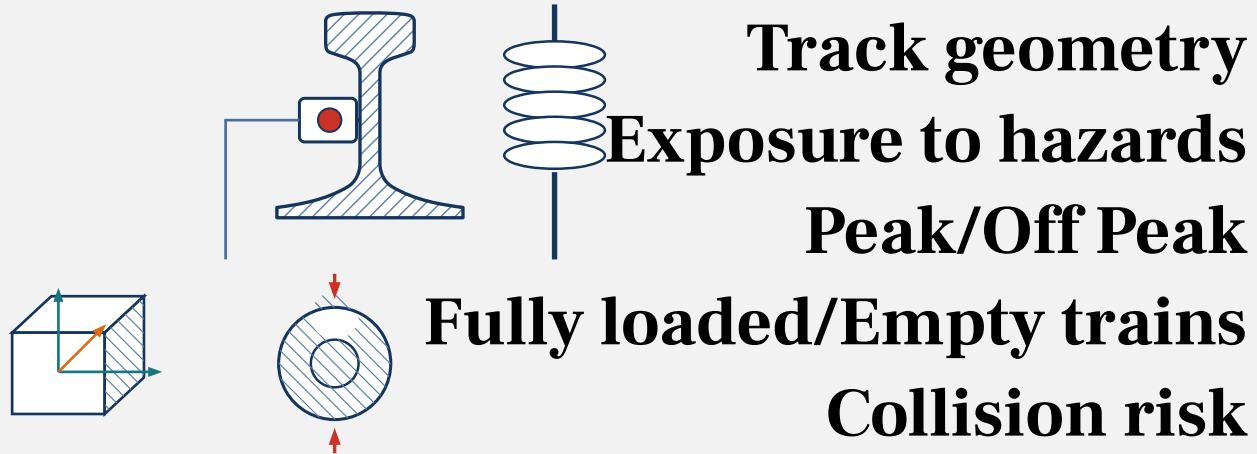


Personnel EQUIPMENT **Third Rail** Evacuation Mass Care



At least 3 additional hazards

1. *track/frog/wheel deterioration*
2. High winds/tradewinds/storms
3. High heat (thermal expansion/buckling)





Other hazards: planning, management, governance

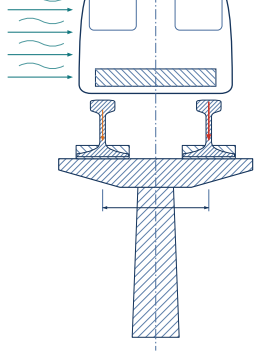
- Purchasing *rail before the vehicle*
- Changing from **2 to 4 car** trains
- Change orders and law suits
- “*Under budget and on-time*”
- Reliance on consultants

• Politics

- Leadership

- *Managing uncertainty*

\$\$\$



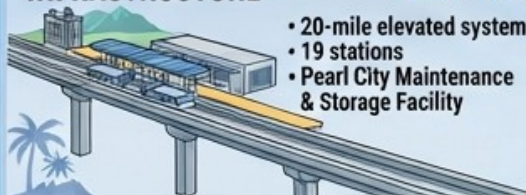
HONOLULU SKYLINE: LOGISTICS, EQUIPMENT, PERSONNEL, & CAPABILITIES.



LOGISTICAL REQUIREMENTS

INFRASTRUCTURE

HART



- 20-mile elevated system
- 19 stations
- Pearl City Maintenance & Storage Facility

POWER SUPPLY



- 750V DC Third Rail Power
- Redundant Grid Connection

SUPPLY CHAIN



- Spare Parts Inventory
- Rolling Stock Maintenance

TRAFFIC MANAGEMENT



- Feeder Bus Connections
- Park & Ride Facilities



SPECIALIZED EQUIPMENT

ROLLING STOCK



- AnsaldoBreda/Hitachi Rail Italy Driverless Metro
- High-Capacity Cars
- 120-person capacity per car

MAINTENANCE VEHICLES



- Hi-Rail Maintenance Trucks
- Track Geometry Measurement System
- Heavy Rescue Road-Rail Excavator

TRAIN CONTROL (CBTC)



- Communications-Based Train Control (CBTC) System
- Automated Operation (GoA4)
- Redundant Data Network

RESCUE TOOLS



- Hydraulic Spreader/Cutter (Jaws of Life)
- Rerailing Equipment
- High-Angle Rescue Kits



PERSONNEL & TRAINING

OPERATIONS TEAM

HART



- Control Center Operators
- Systems Technicians
- Dispatchers

MAINTENANCE CREW



- Rolling Stock Mechanics
- Track & Guldeway Maintainers
- Electromechanics

SECURITY & SAFETY



- HART Security Personnel
- HPD Transit Division
- Emergency Responders Liaison
- K-9 Units

TRAINING



- CBTC Simulation Training
- Emergency Evacuation Drills
- Safety Certification Programs



CAPABILITIES & FUNCTIONS

PASSENGER SERVICE



- 3-minute headways (peak)
- Passenger Info Systems (PIS)

SYSTEM MONITORING



- Real-Time Train Tracking
- Video Surveillance (CCTV)
- Environmental Monitoring

MAINTENANCE & REPAIR

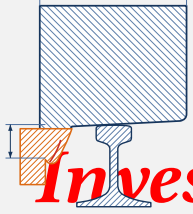


- Preventive Maintenance
- Corrective Repairs
- Component Overhaul

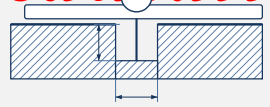
INCIDENT RESPONSE



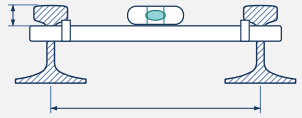
- Automated Train Stoppage
- Passenger Evacuation (elevated & tunnel)
- Disaster Preparedness



Invest in research and development, training, education



FRA

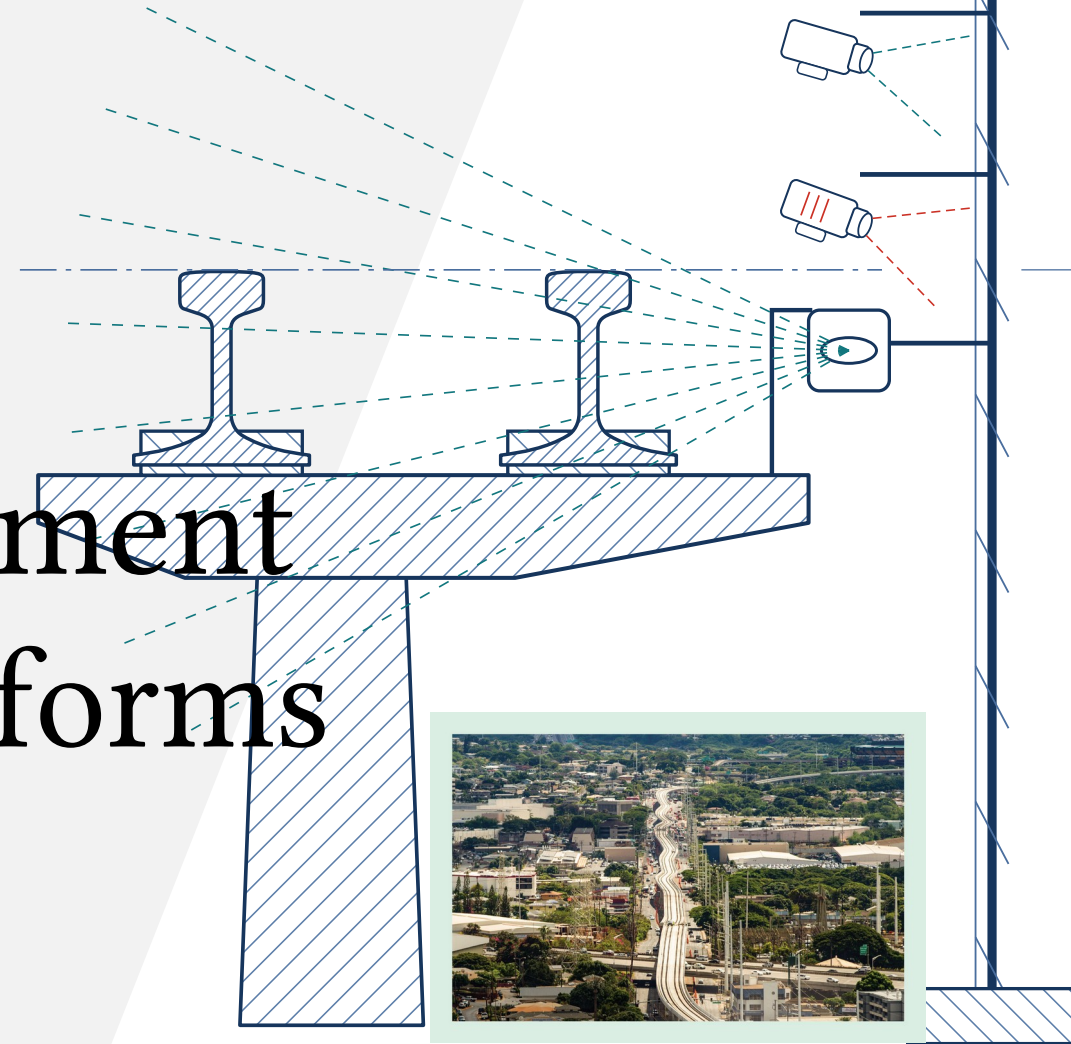


BAA

CRISI

sensor development
integrated platforms
digital twins

HART Skyline; testing training
site





Let's work together

- It's more than **WRI!!!**
- **Design defects and other flaws in**
Planning Engineering Safety Management
Prevent Worst Case Scenarios
INVEST in technology
Communicate, Coordinate, Collaborate
Share knowledge, expertise, experience
LEARN from each other



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SEE YOU SOON!